

Layer 1: Data Ingestion Layer

Ingestion Service (FastAPI:8003) | JWT Authentication | Kafka Publishing

Layer 2: Stream Processing Layer

Kafka + Zookeeper | Spark Master + Worker | Streaming + Aggregations (1min, 5min, 1hr)

Layer 3: Storage Layer

TimescaleDB (PostgreSQL 15:5432) | Redis Cache (6379) | MinIO Object Storage (9000)

Layer 4: MLOps Layer

MLflow Tracking (5000) | Jupyter (8888)
ML Service API (8002) | MinIO Model Registry

Layer 5: Application Layer

API Gateway (8000) | Alert API (8001)
Frontend (3000) | ML Service (8002)
Monitoring: Prometheus, Grafana, Loki